—Everettian many-worlds, eternal inflation, and the string landscape—were in fact a single multiverse, and the three theories were simply describing the same reality in different ways. In any case, the many-worlds interpretation was (mostly) no longer laughed out of the room without serious consideration. Indeed, by the start of the twenty-first century, many-worlds had become the most popular rival to the Copenhagen interpretation itself among physicists, with particular popularity among cosmologists. But with wider consideration came the realization of a new problem, one that any theory with an infinite multiverse faced: the problem of probability.

At its core, the measurement problem asks when wave functions obey the deterministic harmony of the Schrödinger equation and when they undergo the random process of collapse. The many-worlds interpretation gets around the measurement problem by denying that wave function collapse happens at all. In the multiverse of many-worlds, the wave function of the universe always obeys the Schrödinger equation, splitting off into an endless series of branches that constitute the many worlds. But there's a problem with this picture. It's not clear how randomness and probability enter our quantum physics experiments if the universal wave function really does obey the Schrödinger equation at all times, since that equation is completely deterministic, with no element of chance. After all, one thing that absolutely everyone agrees on, no matter what interpretation (or incoherent pseudo-interpretation) they subscribe to, is that the outcomes of quantum physics experiments have an element of randomness. In general, the mathematical machinery of quantum physics only lets us predict the probabilities of particular experimental outcomes, rather than stating with certainty that a particular thing will happen. But if the entire universe is deterministically obeying a single equation, how can probability enter into physics at all?

Normally we think of probability as being like rolling a die: there are six outcomes, you can only get one, so the odds of any particular outcome are one in six (unless the die is weighted). The probability of rolling an odd number on a die is three in six, because there are three ways to get that outcome out of six total possible outcomes (Figure 11.1a). But probability